

ECEN3017 Analog Integrated Circuit Design

Project-3

Fully Differential Two-Stage Miller Op-Amp Design

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Abstract—Combining the knowledge learned through the course, this project aims to design a fully differential two-stage operational amplifier (Opamp) based on 55nm processing, which incorporates a two-stage core with Miller compensation, a current mirror-based biasing circuit, and a common mode feedback (CMFB) circuit to ensure stability and optimal performance. Key performance metrics were carefully analyzed through detailed calculations and simulations to successfully meet the specifications. Finally, the simulation results demonstrate that the amplifier consumes around 6.73 mW to achieve 74.67 dB DC gain, 1.55G Hz gain bandwidth product (GBW), 69.78° phase margin (PM), and -69.3 dB total harmonic distortion (THD) with 1.03V/ns slew rate (SR) for the closed-loop testing.

Index Terms—Fully differential two-stage Miller op-amp, Telescopic amplifier, Common source (CS) amplifier, Current mirror-based biasing circuit, Common mode feedback (CMFB), Miller compensation.

I. INTRODUCTION

THE two-stage amplifier is a structure designed to amplify signals through two distinct stages, thus to achieve significantly better performance than the single-stage design. Each stage serves specific purpose. Normally, the first (main) stage enables higher gain and the second stage provides a larger output swing.

However, multi-stage also introduces oscillations caused by high-frequency poles. A two-stage amplifier has a low-frequency dominant pole at the first-stage output and a higher-frequency non-dominant pole at the second-stage output. Both poles locate at the left-half-plane (LHP pole), which will cause 90-degree phase drop each. In this case, when the frequencies of two poles are close to each other, a sharp decline of 180 degrees may happen within the bandwidth, leading to instability. Therefore, Miller compensation is used to split the two poles and control the bandwidth by connecting a capacitor between the outputs of the first and second stage. A resistor is also added in series with the Miller capacitor to eliminate the right-half-plane zero (RHP zero) that introduced by the Miller capacitor. By pushing backward the first pole to a lower frequency within the bandwidth while pushing towards the second pole to a

higher frequency within 2 to 3 times of the bandwidth, Miller compensation can greatly improve the stability of the whole system.

Moreover, unlike single-ended amplifiers where the output DC level is fixed by biasing, a fully differential amplifier has two branches with floating outputs. Mismatches in transistors, resistors, or capacitors will cause imbalance between the two branches, making the common-mode output voltage (V_{CM}) sensitive to process variations and load changes, corrupting the differential signal. Therefore, the common-mode feedback is necessary to actively monitor and regulate V_{CM} to V_{REF} set to stabilize the DC voltage of the differential output nodes, ensuring maximal differential swing.

Given this situation, the basic circuit structure is determined as shown in Figure 1 and the performance specifications are also shown below in Figure 2. The overall design procedure will be separated into whole structure consideration, amplifier design, CMFB design, biasing circuit design, Miller compensation, and close loop testing as shown in the following report.

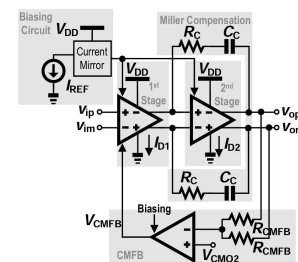


Fig. 1. The fully differential two-stage Miller Op-Amp circuit structure.

Voltage Supply	$V_{DD}=1.2V$
DC Gain	$A_0 > 55dB$ (>70dB)
GBW	$> 500MHz$ (>1GHz)
Phase Margin	PM > 62°
Closed-loop Output Swing	$V_o > 1.2V_{pp}$ differential @ 1MHz with THD < -40dB (>1.8Vpp differential, THD < -65dB)
Slew Rate	SR > 0.5V/ns (>1Vns)
Loading	$Z_L = C_L = 1pF$
Power Consumption	< 10mW (the lower, the better)

** Achieving the specification in brackets indicates the bonus.

Fig. 2. Performance specifications.

II. WHOLE STRUCTURE CONSIDERATION

The consideration of voltage, current, and other critical parameters distribution is very important before the design and simulation, which highly determines whether the specifications can be reached. For this project, there are mainly two points worthy of attention: the structural selection for 3 amplifiers, and the calculations for all critical parameters.

A. Structural Selection for 3 Amplifiers

As mentioned above, the first stage of the amplifier contributes the main gain, and the second stage of the amplifier provides a large output swing. To satisfy the requirement of the overall gain higher than 70 dB, the fully differential telescopic structure as shown in Figure 3 is chosen as the first stage, which can provide a gain up to 48 dB with a power consumption of 1 mW according to the experience of project 2. The telescopic structure achieves a large output resistance that leads to a higher gain, which satisfies the equation (the number is the same as Figure 3):

$$A_v = -g_{m1}[(g_{m3}r_{o1}r_{o3})||((g_{m5}r_{o5}r_{o7}))] \quad (1)$$

However, with more MOSFETs in each branch, the output swing will be compressed. Therefore, a simple fully differential common-source (CS) structure is used for the second stage as shown in Figure 4, which can provide a larger output swing for the whole circuit. The gain of the second stage, which provides around 25 dB is enough, can be calculated as (the number is the same as Figure 4):

$$A_v = -g_{m1}(r_{o1}||r_{o3}) \quad (2)$$

A MOSFET used as the current source to maintain the current flow through the amplifier is used in both stages. With a larger output impedance seen at the common-source node, it can also provide a better common-mode rejection ratio (CMRR), which suppresses common-mode noise at the input end to ensure pure differential signal amplification.

In addition, the voltage distribution and the settings of terminal voltages for each MOSFET on the branch need to be carefully chosen. To maximize the output swing, it's better to set the output DC voltages V_{out} of both stages to be half of the voltage supplied (V_{DD}), which equals 600 mV. In this case, other voltage settings can be easily derived by letting every MOSFET saturate with $V_{dsat} \approx V_{ov} = V_{GS} - V_{TH}$ within the range from 100 mV to 200 mV. However, it's difficult to meet the V_{dsat} requirement for the differential pairs that need high g_m to satisfy the gain specification. With the same current setting in the branch, the V_{dsat} of the differential pairs is usually set less than 100 mV to have enough g_m with a much larger width, where:

$$g_m = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right) I_{DS}} = \frac{2I_{DS}}{V_{GS} - V_{TH}} \quad (3)$$

Also, other MOSFETs such as M3 and M5 may have a larger

length to provide larger output impedances:

$$R_o = \frac{1}{\lambda I_{DS}} \quad (4)$$

Where λ is inversely proportional to the length L .

For the CMFB circuit, a diode-connected PMOS in series with an NMOS is normally used as shown in Figure 5. This structure provides little gain that can be expressed as (the number is the same as Figure 5):

$$A_v = -g_{m1} \left(r_{o1} || \frac{1}{g_{m2} + g_{mb2}} \right) \quad (5)$$

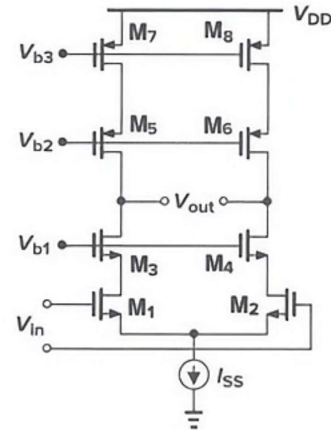


Fig. 3. The fully differential telescopic amplifier.

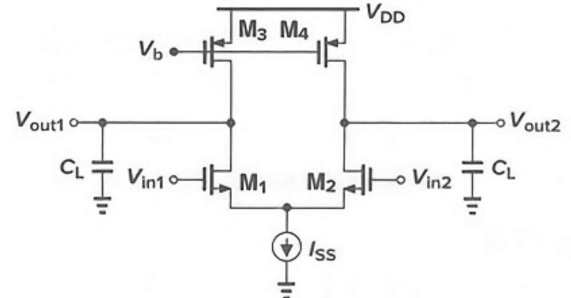


Fig. 4. The fully differential CS amplifier.

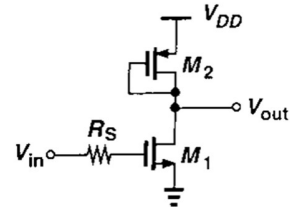


Fig. 5. The amplifier structure used in the CMFB circuit.

B. Calculations

As the specifications indicate that $GBW > 1\text{GHz}$, $PM > 62$ degree, $THD < -65\text{dB}@1.8\text{Vpp}$ differential, and $SR > 1\text{Vns}$. These critical parameters should be roughly calculated before the simulation, which can be expressed as follows:

$$GBW = \frac{g_{m1}}{2\pi \times C_c} \quad (6)$$

$$\omega_{p1} \approx -\frac{1}{R_1[C_1 + (1 + g_{m2}R_2C_c)]} \approx -\frac{1}{R_1(g_{m2}R_2)C_c} \quad (7)$$

$$\omega_{p2} \approx -\frac{g_{m2}C_c}{C_1C_c + C_2C_c + C_1C_2} \approx -\frac{g_{m2}}{C_2} \quad (8)$$

$$SR = \frac{I_{SS}}{C_c} \quad (9)$$

Where g_{m1} , g_{m2} , C_1 , and C_2 represent for the whole stage, not a single MOSFET. C_c is the capacitance of the Miller capacitor. Initially, set the total current flow through the first stage I_{SS} to be 800 μA and the g_{m1} to be 8 mS according to the experience of Project 2. Therefore, to satisfy the specifications, calculate the parameters with some margins:

$$GBW = 1.3G = \frac{8m}{2\pi \times C_c} \quad (10)$$

$$SR = 1.2G = \frac{800\mu}{C_c} \quad (11)$$

C_c should be smaller than 980 fF according to Equation 9 and it should be smaller than 666.7 fF as well according to Equation 10. Also, the frequency of the two poles can be calculated as below, referring to Equations 6 and 7:

$$f_{p1} \approx -\frac{1}{2\pi R_1[C_1 + (1 + g_{m2}R_2C_c)]} < 1G \quad (12)$$

$$3G > f_{p2} \approx -\frac{g_{m2}C_c}{2\pi(C_1C_c + C_2C_c + C_1C_2)} > 2G \quad (13)$$

In this case, g_{m2} needs to be high enough to satisfy these relationships. Furthermore, the location of the RHP zero and the resistance to cancel it can also be obtained as:

$$f_z = \frac{g_{m2}}{2\pi C_c} \quad (14)$$

$$R_z = \frac{C_c + C_2}{C_c g_{m2}} \quad (15)$$

III. AMPLIFIER DESIGN

A. First-Stage Telescopic Amplifier

After the calculation, the first-stage amplifier is designed as shown in Figure 6. I_{SS} is set to be 800 μA , thus the power consumption is calculated to be 960 μW . The g_m of the differential pair is 8.31m, which can be roughly considered as the g_m for the first stage according to Equation 1. The size settings for each MOSFET are shown in Table 1 below, which follow the rules mentioned above to get a larger gain (M1&2),

output impedance (M3&4, M5&6, M7&8), or better CMFB performance (M9).

Finally, the telescopic amplifier achieves a gain = 51.818 dB with a bandwidth = 59.6 MHz as shown in Figure 7. The DC input voltage is as same as the common mode voltage = 600 mV and 3 voltage biases are needed for the amplifier, where $V_{BN1} = 900$ mV, $V_{BP1} = 250$ mV, and $V_{BP2} = 600$ mV.

TABLE I. SIZE SETTINGS FOR THE FIRST-STAGE TELESCOPIC AMPLIFIER

	M9	M1&2	M3&4	M5&6	M7&8
Length	250n	100n	150n	150n	150n
Total Width	70.8u	154.596u	49u	51.496u	58.896u
Fingers	8	12	8	8	8
Finger width	8.85u	12.883u	6.125u	6.437u	7.362u

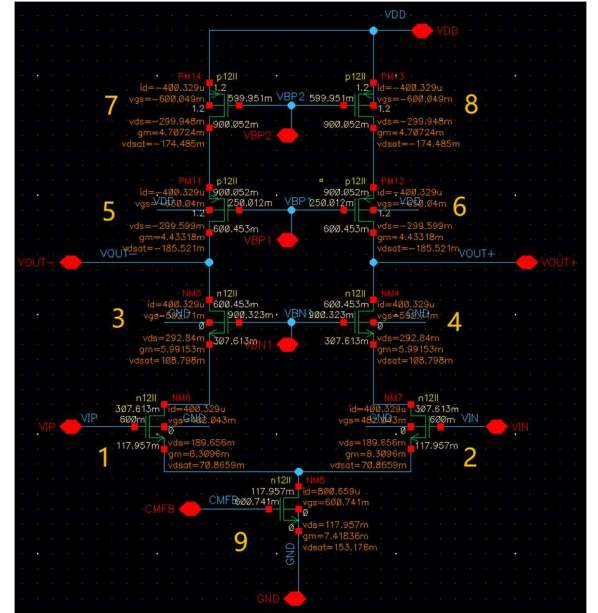


Fig. 6. The first-stage telescopic amplifier.

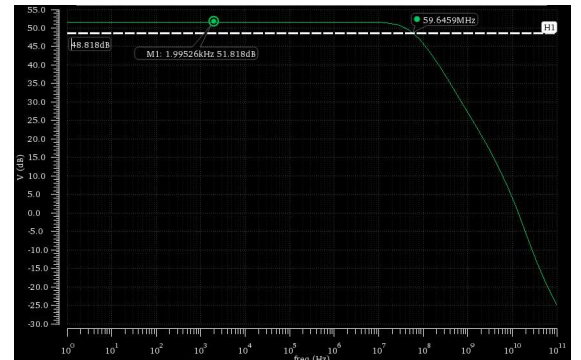


Fig. 7. The gain and bandwidth of the first-stage telescopic amplifier.

B. Second-Stage CS Amplifier

The second-stage amplifier is designed as shown in Figure 8. Total current pass through is set to be 3.6 mA, thus, the power consumption is calculated to be 4.32 mW. The g_m of the

differential pair is 34.2239m, which is quite large to satisfy the requirement for Miller compensation and can be roughly considered as the gm for the second stage according to Equation 2. The size settings for each MOSFET are shown in Table 2 below, which follow the rules mentioned above to get a larger gain (M10&11), output impedance (M12&13), or better CMFB performance (M14).

Finally, the CS amplifier achieves a gain = 22.88 dB with a bandwidth = 1.35 GHz as shown in Figure 9. 1 voltage biases are needed for the PMOS and the tail NMOS of the amplifier, where VBP2 = 600 mV.

TABLE II. SIZE SETTINGS FOR THE SECOND-STAGE CS AMPLIFIER

	M14	M10&11	M12&13
Length	300n	100n	150n
Total Width	227.504u	341.1u	167.0044u
Fingers	16	20	12
Finger width	14.219u	17.055u	13.917u

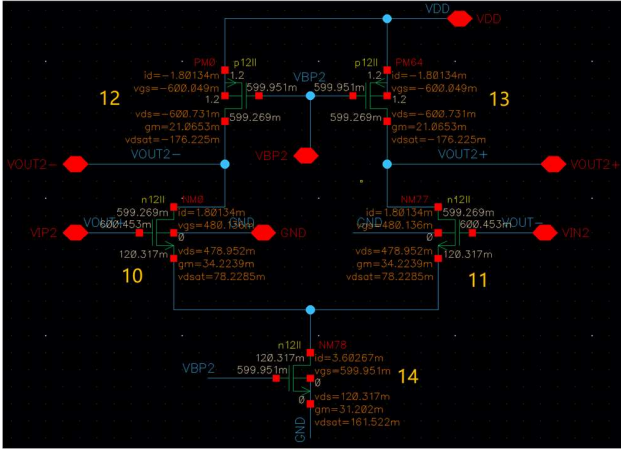


Fig. 8. The second-stage CS amplifier.

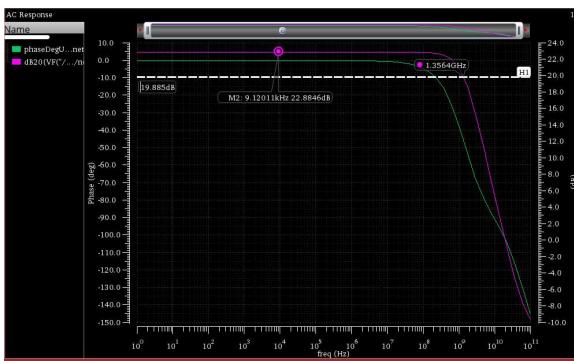


Fig. 9. The gain and bandwidth of the second-stage CS amplifier.

IV. CMFB DESIGN

An ideal CMFB circuit as shown in Figure 10 can be built first to check whether the two-stage amplifier works. With a gain = 10, the circuit senses the outputs and then feeds back the voltage:

$$V_{CMFB} = Error \times (-10) + V_G \left[\frac{1}{2} \times (V_{OUT+} + V_{OUT-}) - V_{CM} \right] \times (-10) + V_G \quad (16)$$

Then the real CMFB circuit is built as shown in Figure 11, which can be calculated to provide a gain = 0.28 dB according to Equation 5, but simulated with a gain = -222.2 m dB with a bandwidth = 2.1 GHz according to Figure 12. The size settings are shown in Table 3. The common mode voltage = 600 mV is compared with the second-stage output voltage sensed at the input and then fed back to the first-stage amplifier.

One thing should pay attention to is the resistance chosen for the sensing resistor. It can't be too large or too small. No current will flow through the CMFB circuit so that no feedback will be output if the resistance is large enough as 100 MΩ, while a large amount of current will flow in so that will cause damage to the two-stage amplifier if the resistance is too small as 10 Ω. The setting of 100 KΩ is a good choice for this project.

TABLE III. SIZE SETTINGS FOR THE AMPLIFIER FOR THE CMFB CIRCUIT

	M22	M18&19	M20&21
Length	400n	400n	100n
Total Width	110.496u	102.396u	150.804u
Fingers	8	12	12
Finger width	13.812u	8.533u	12.567u

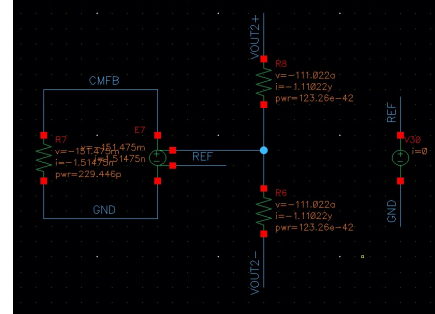


Fig. 10. The ideal CMFB amplifier.

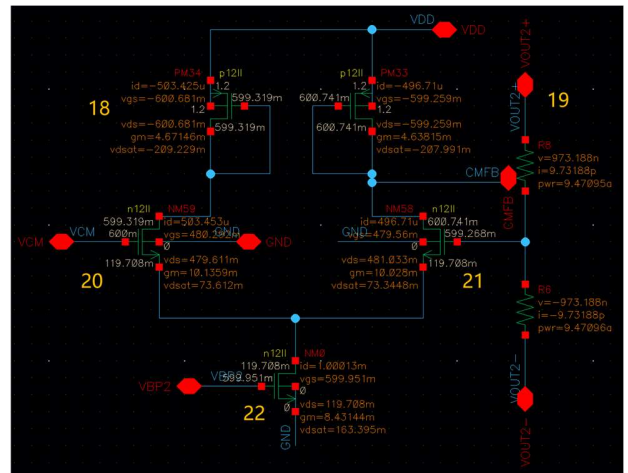


Fig. 11. The real CMFB amplifier.

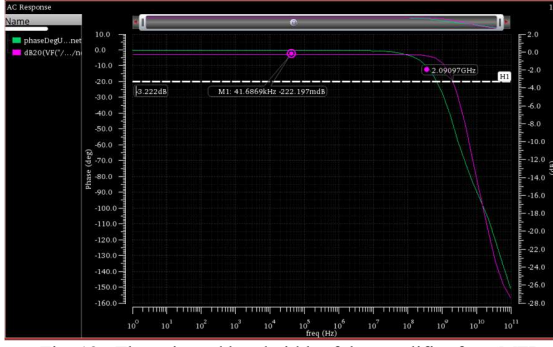


Fig. 12. The gain and bandwidth of the amplifier for CMFB.

V. BIASING CIRCUIT DESIGN

With the experience in project 2, the current-mirror-based biasing circuit is used to stabilize the operating point of the transistor. It can reduce the packaging cost by reducing additional pins for DC bias voltage sources as well. Provide a reference current equal to one-eighth of the current (50 uA) in a single branch of the first-stage amplifier. Then copy the current to each branch by letting the last NMOS at each branch has a similar size to M15. Therefore, the sizes of M15, M16, and M17 can be determined as shown in Table 3, which will have one-eighth of the width of M7, M5, and M3 accordingly. The other MOSFETs' sizes can also be easily obtained to let it saturate and maintain the branch current.

TABLE IV. SIZE SETTINGS FOR THE BIASING CIRCUIT

	M15	M16	M17
Length	150n	150n	150n
Total Width	4.55u	6.437u	6.125u
Fingers	1	1	1
Finger width	4.55u	6.437u	6.125u

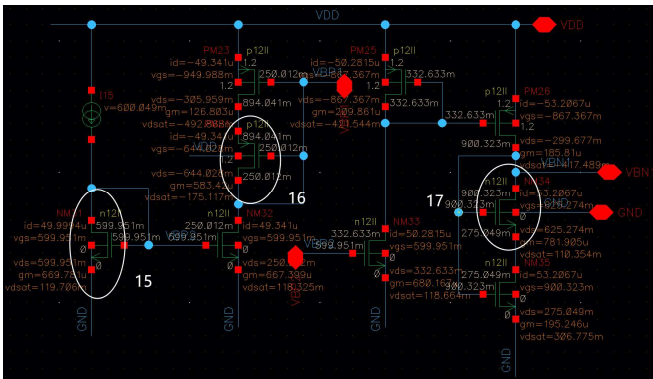


Fig. 13. The current-mirror-based biasing circuit.

VI. MILLER COMPENSATION

According to Miller's theory as shown in Figure 14, when applying a negative feedback gain, the equivalent capacitance at the first stage $C_{eq1,1}$ and the equivalent capacitance at the second stage $C_{eq1,2}$ can be expressed as:

$$C_{eq1,1} = C_C(1 + A) \approx AC_C \quad (17)$$

$$C_{eq1,2} = C_C(1 - 1/A) \approx C_C \quad (18)$$

In this case, by increasing the capacitance in the output of the first stage and remaining the capacitance in the output of the second stage nearly the same. The split of the two poles can be achieved.

As the critical parameters are obtained, $gm1 = 8.31m$, $gm2 = 34.2m$. According to Equations 10 and 11, let the Miller capacitor to be 650 Pf. Also, the capacitance can be calculated:

$$C_{tot} = C_C + C_{dg10} = 650f + 79f = 749f \quad (19)$$

$$C_1 = C_{dd4} + C_{dd6} + C_{gg10} = 10.3f + 12.2f + 323.6f = 346.1f \quad (20)$$

$$C_2 = C_L + C_{dd10} + C_{dd12} = 1p + 78.8f + 52.4f \approx 1.1p \quad (21)$$

Therefore, the resistance connected in series and the frequency of the second LHP pole can be calculated according to the Equations 15 and 13:

$$R_z = \frac{749f + 1.1p}{749f \times 34.2m} = 73.35\Omega \quad (22)$$

$$f_{p2} \approx \frac{34.2m \times 749f}{2\pi \times (346.1f \times 749f + 1.1p \times 749f + 346.1f \times 1.1p)} = 2.79G \quad (23)$$

In this case, the second pole is properly pushed forward to the frequency range from 2G ~ 3GHz. However, the calculation result is not quite accurate. By scanning the phase margin with the change of R_z , when R_z is around 150Ω , the PM will have the optimal performance to be 69.78° . PM should not be too large or too small. The system will be unstable if PM is smaller than 45° , while the response time will get worse if PM is larger than 90° .

Therefore, the Miller capacitance and the resistance are finally determined as:

$$R_z = 150\Omega$$

$$C_C = 650pF$$

The Bode plot before and after the Miller compensation are also shown as a comparison below in Figure 15 and 16. Moreover, the open-loop specifications are all achieved as shown in Figure 17, which are similar to the calculation results.

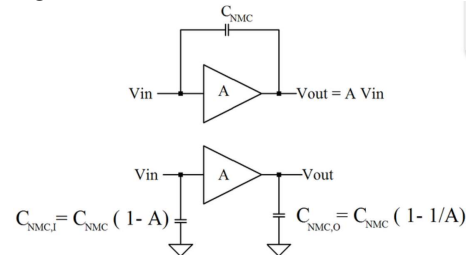


Fig. 14. The Miller theory.

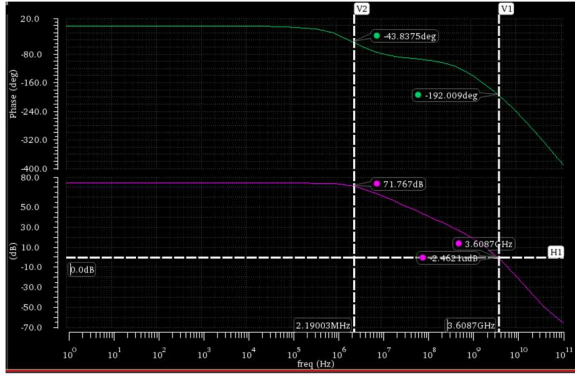


Fig. 15. The Bode plot before Miller compensation (phase in green and gain in pink; PM = -12deg).

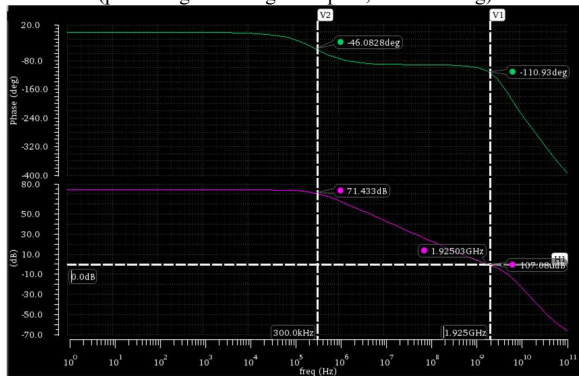


Fig. 16. The Bode plot after Miller compensation (phase in green and gain in pink; PM = 69.07deg).

Outputs	
Name/Signal/Expr	Value
1 Power	-6.72755m
2 PM	69.7772
3 Gain	74.6663
4 GBW	1.55252G

Fig. 17. The simulation results for the open-loop test.

VII. CLOSED-LOOP TESTING

The circuit connections of open-loop and closed-loop are shown in Figures 18 and 19. By connecting the feedback resistors 1K and 5K Ω , the closed-loop gain is 5. As shown in Figures 20 and 21, the simulation results of closed-loop testing satisfy the specification, where SR achieves 1.03V/ns and THD = -69.3 dB @ 1MHz with >1.8Vpp differential (where the amplitude of the input sinusoidal wave is 180 mV).

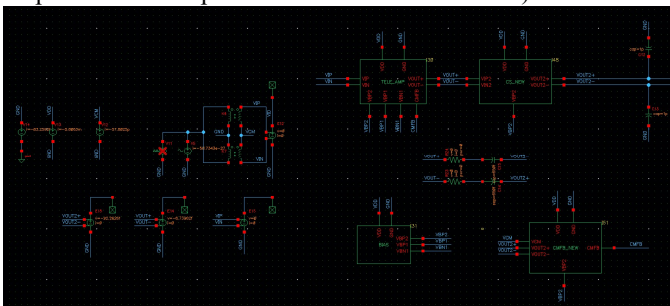


Fig. 18. The open-loop circuit.

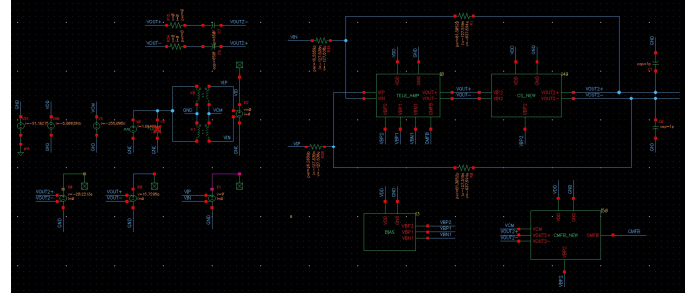


Fig. 19. The closed-loop circuit.

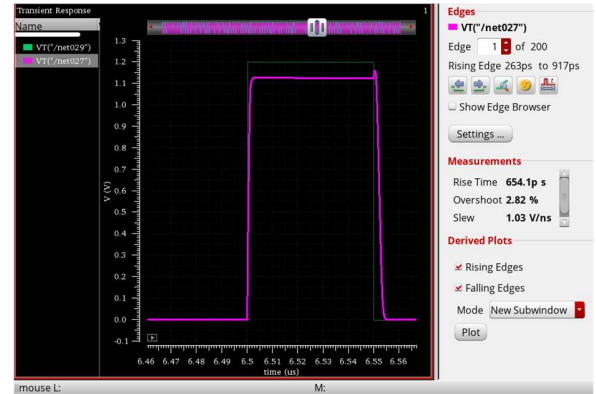


Fig. 20. The simulation result of SR.

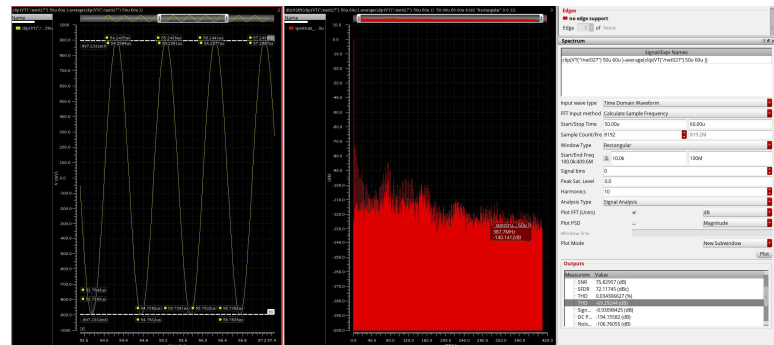


Fig. 21. The simulation result of THD.

VIII. DISCUSSIONS & IMPROVEMENTS

During the whole design process, many improvements and changes are made to get a better performance:

To begin with, an operational transconductance amplifier that contains 5 MOSFETs (5-OTA) as shown in Figure 22 is originally used in the CMFB circuit, which can provide a relatively large feedback gain up to 20 dB. Though a larger feedback gain can provide a greater matching to the differential amplifier, it can also increase the loop gain of the CMFB loop, pushing its phase margin toward 0° and risking oscillations, which can be clearly shown by the transient signal when doing the SR testing according to Figure 23. The feedback gain is not only provided by this amplifier as well. In this case, a diode-connected PMOS with another NMOS in series as applied in this project is more often used as the amplifier structure for the CMFB Circuit.

In addition, the setting of the gm for the second stage is also

quite considerable, which is adjusted from 10m, 25m, and finally to 34m to achieve the SR specification. It's really important to calculate with some margins before designing and the simulations.

In conclusion, the two-stage amplifier designed in this project successfully achieves all the specifications. However, there is still much that can be improved.

Firstly, the length of M3&4 and M5&6 in the first-stage amplifier could be at least doubled in size to provide a larger output impedance.

Also, the MOSFET M9 that acts as the current source for the second-stage amplifier is usually not used in a commonly designed two-stage amplifier. Although it can provide better CMRR performance, the MOSFET also lowers its output swing.

Moreover, instead of reducing the size of the Miller capacitor to achieve the requirement of SR, which then leads to a significant increase in the requirements for the second-stage g_m and power consumption to maintain the Miller compensation, a better choice is to increase the I_{SS} in the main stage according to Equation 9. Remember to larger the second-stage current simultaneously.

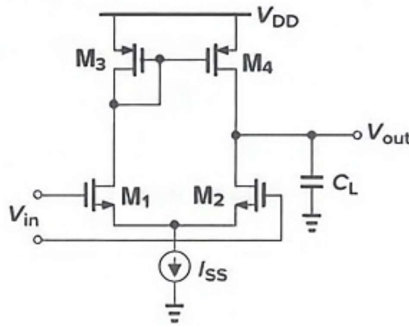


Fig. 22. 5-OTA for the CMFB circuit.

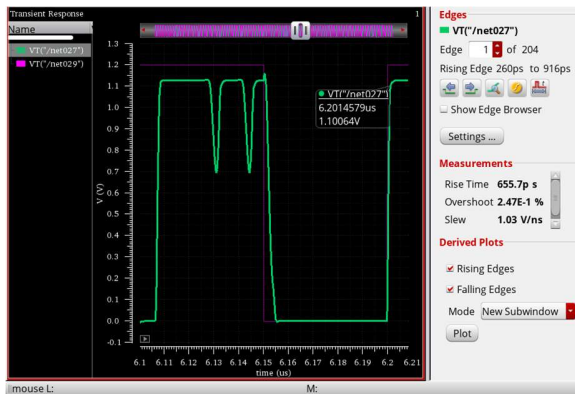


Fig. 23. The oscillated transient signal due to the large CMFB gain.

IX. CONCLUSION

In conclusion, the fully differential two-stage operational amplifier designed in this project successfully meets all the specified performance requirements as shown in Table 5 while operating in a 55nm CMOS process. The key to completing this project is learning trade-offs. For instance, trade-offs between power and SR, PM and SR, and power and gain. The optimal

performance will be obtained by continuous calculation and simulation.

This final project significantly enhanced the author's understanding of analog circuit design. With practical experience in both hand calculation and simulation, the project bridged the gap between theory and implementation, solidifying the author's skills in designing and optimizing high-performance analog circuits.

TABLE V. OVERALL PARAMETERS ACHIEVEMENTS

Supply Voltage	1.2V
DC Gain	$A_o=74.7\text{dB}$
Gain Bandwidth Product	$\text{GBW}=1.55\text{GHz}$
Phase Margin	$\text{PM}=69.78^\circ$
Closed-loop Output Swing	$>1.8V_{pp}$ differential @1MHz THD = -69.3dB
Slew Rate	1.03V/ns
Loading	$C_L=1\text{pF}$
Power Consumption	6.73mW

X. ACKNOWLEDGEMENT

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